



MADHAVA BRIGHTLY A

PYTHON BACKEND DEVELOPER

<https://madhavabrightly.github.io/portfolio-main/#projects>

Contact

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Education

(2022 - 2026) **ANNAMALAI UNIVERSITY**
BE-INFORMATION TECHNOLOGY

LANGUAGE

TAMIL
ENGLISH
LATIN-BEGINNER

Skills

- Python, JavaScript
- AI/ML (YOLO, Deep Learning)
- FinTech Simulation Systems
- Linux Administration
- Figma (UI/UX Design)
- Problem Solving & Algorithms

Current Work

- Engineered automated testing frameworks to monitor system hooks, network interfaces, and audio hardware routing under virtualization to analyze mobile system behaviors.
- Mastering vanilla JavaScript and native runtime APIs on the fly to build optimized data extraction pipelines and real-time scrapers for AI workflows.
- Managed a successful technical freelancing portfolio, delivering custom full-stack web, mobile, and AI applications for client and academic requirements.
- Structured a personal development routine balancing advanced technical research with wellness habits to maximize cognitive performance and prevent burnout.

About Me

IT professional specializing in multi-model prompt engineering pipelines that connect high-level system architecture with granular code generation to accelerate development. Adept at translating technical requirements into structured LLM context to automate data workflows, streamline debugging, and rapidly deploy secure web, mobile, and AI applications.

PROJECTS

2026

SYNTHETIC MARKET

Developed a synthetic market simulation web application enabling users to practice trading strategies in a risk-free environment, incorporating realistic market dynamics, advanced order execution logic, and a hybrid trading bot combining rule-based and algorithmic strategies with a 40% target profit model and 20% automated stop-loss/auto-close mechanism to simulate disciplined trading behavior

2025

FACIAL RECOGNITION USING AI

Developed an AI-based facial emotion recognition system that detects and classifies human emotions from facial expressions using computer vision and deep learning techniques, enabling accurate real-time analysis

2024

PYTHON SCREEN RECORDER

Developed a crash-resilient Windows desktop recorder utilizing FFmpeg to capture stable 60 FPS video alongside a dual-channel microphone and system loopback audio mixer. Engineered real-time hardware hot-swapping to adapt to audio device changes mid-recording, integrating spectral grating filters and low-frequency rumble removal to eliminate background noise.

Working Experience

Software Engineer with 2+ years of freelance experience and professional experience as a Full Stack Developer and Generative AI/LLM Engineer, specializing in AI-powered applications, backend APIs, and modern web development.